

Brain Tumor Segmentation Based on Improved Swin-UNet

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Abstract. Swin UNet combines the characteristics of Swin Transformer and UNet. UNet is a classic image segmentation network with an encoder decoder structure that can provide contextual information while preserving spatial information. Swin Transformer provides a cross window attention mechanism that allows models to obtain global information on images. The core idea of Swin Transformer is to divide the image into non overlapping blocks, and then introduce a cross window attention mechanism, allowing the model to perform self attention calculations within each window to capture global and local information of the image. Swin UNet is mainly used for medical image segmentation tasks, especially for tumor segmentation in brain MRI images. A new segmentation model is proposed by combining the global information acquisition ability of Swin Transformer and the encoder decoder structure of UNet. Swin-UNet-EPA aims to improve segmentation accuracy and performance. For brain tumor segmentation tasks, the performance evaluation of the model usually uses indicators such as IoU and Dice coefficient to measure the degree of matching between the predicted results of the model and the actual labels.

Keywords: unet;swin-unet; brain tumor segmentation; deep learning.

I. Introduction

Brain tumors are one of the most common diseases that seriously endanger human life and health. Malignant tumors such as gliomas are prone to recurrence and have a high mortality rate. [1] Machine learning is one of the fastest-growing branches in the field of artificial intelligence. Utilizing the computing speed advantages of high-performance computers, it can quickly and comprehensively screen important information from complex data dimensions and massive data content, helping humans obtain general laws. [2] Applying machine learning to the field of brain tumor segmentation is expected to improve the accuracy of model automatic segmentation. This is necessary and helpful for clinical medicine. For brain MRI, there are four commonly used multimodal images: FLair images, T1 weighted images, T1ce images, and T2 weighted images. MRI images of different modalities have different imaging effects on different tumor regions [3]. Currently, many

popular deep neural networks have been widely used in the field of image segmentation, such as Unet [4], AlexNet[5], SegNet [6], ResNet [7], and so on. The main method for brain tumor image segmentation based on deep learning is the fully convolutional neural network proposed in 2012 [8]. The Unet model is a very classic model, while Swin Unet is a deep learning network that combines Swin Transformer and U-Net architecture for image segmentation tasks.[9]

II. EPA

As shown in Figure 1. The EPA is an efficient paired attention block based on spatial and channel attention. In traditional self attention, each position's Query, Key, and Value has its own parameter matrix, resulting in a rapid increase in computational complexity as the sequence length increases. In EPA, it adopts a parameter sharing approach, where the Query, Key, and Value parameters are the same at all positions, thereby reducing the number of parameters. The two attention modules of EPA share keys and queries, which can be passed on to each other and generate more effective feature representations. In the attention mechanism, the similarity (dot product) between Query and Key is calculated, and the results are normalized to attention weights to determine which parts of the input sequence should receive more attention at the current time step. These weights will be used to weight the values in the input sequence to generate the final attention pool, which can then be used in conjunction with information from other parts to generate output. Key and Query work together in the attention mechanism to help the model decide which information to focus on when processing input image segmentation. The EPA also calculates attention weights through a balanced weighting approach, ensuring that information at each location is given appropriate attention. This is different from traditional self attention mechanisms, which may lead to overly focused attention in certain positions while ignoring others. At present, the computational complexity of the self attention mechanism can reach quadratic, and its computational complexity is very complex.

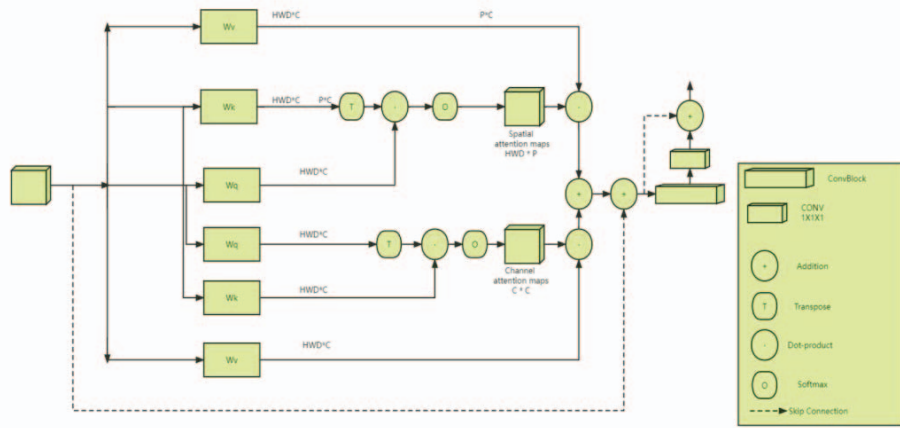


Figure 1. EPA

III. Swin-Unet

In deep learning, Transformer ^[10] is an architecture originally proposed by Vaswani et al. in 2017, mainly used for natural language processing tasks such as machine translation. However, due to its excellent performance and versatility, Transformer has also been successfully applied in other fields, including computer vision. The self attention mechanism is a core component of Transformer. Allow the model to weight information at different positions in the sequence when processing sequence data. This enables the model to consider the context of the entire sequence while processing each element. In order to improve the expressive power of the model, Transformer uses multiple parallel self attention mechanisms, each referred to as a head. Each head learns different weights

and finally concatenates their outputs together. After the attention layer, each position has a feedforward neural network. This is a fully connected feedforward structure used for nonlinear transformation of the output of the attention mechanism at different positions. The output of each sub layer is normalized through layer normalization and then added to the input (residual connection). This helps to alleviate the gradient vanishing problem during the training process and simplifies the training of the model. The Transformer does not contain convolutional layers, so it cannot capture the position information of elements in the sequence. In order to introduce positional information, positional encoding is added to the input embedding, allowing the model to consider the relative position of elements when processing sequences.

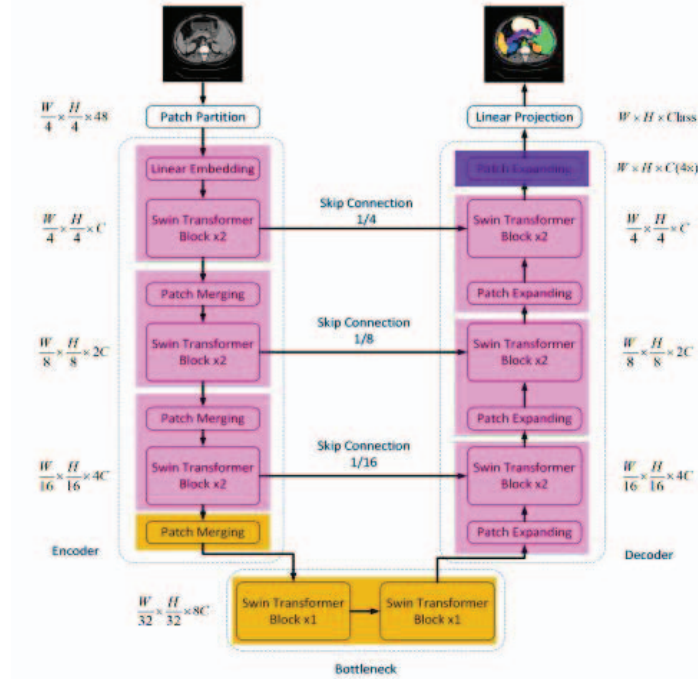


Figure 2. Swin-Unet

As shown in Figure 2. Swin Transformer is a variation of the traditional Transformer architecture designed for image processing tasks. It utilizes layered methods to capture information at different spatial resolutions, making it particularly suitable for computer vision tasks. The Swin Transformer consists of a set of stages, each processing input at different spatial resolutions. On the other hand, U-Net is a convolutional neural network architecture used for tasks such as image segmentation, which combines contraction paths (for capturing context) with extension paths (for precise positioning). Use Swin Transformer as the encoder part of the network. The encoder is responsible for capturing hierarchical features of different spatial resolutions. Each stage in the Swin Transformer can process input images at different spatial scales, enabling it to effectively capture local and global information. Use the U-Net architecture as the decoder part. The decoder obtains layered features extracted by the Swin Transformer and combines them with higher resolution feature maps from the contraction path. This allows the network to perform precise localization and segmentation. Final output layer: Typically, there is a final output layer that generates pixel predictions for segmentation tasks. This output layer can use convolutional layers with appropriate activation functions for multi class or binary segmentation. The combination of Swin Transformer as an encoder and U-Net as a decoder can leverage the advantages of these two architectures. Swin Transformer excels at capturing contextual and global information, while U-Net is specifically designed for precise positioning, making it a potentially powerful architecture for image segmentation tasks.

IV. Experimental results and analysis

A. Datasets

The 2021 BraTs dataset was used, with 1000 pieces of data as the training set, 125 pieces of data as the validation set, and 125 pieces of data as the test set. From figure 3, BraTS contains four modalities of magnetic resonance imaging per case, with each modality having a dimension of $240 \times \text{two hundred and forty} \times 155$ ($L \times W \times H$).

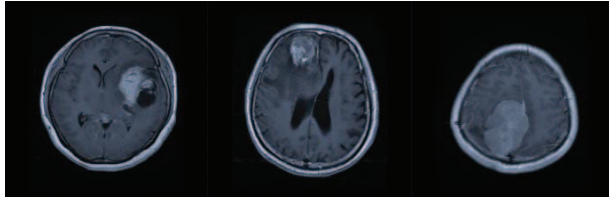


Figure 3. BraTs Dataset

B. Evaluation indicators

$$\text{DICE} = \frac{2(X \cap Y)}{X + Y} \quad (1)$$

As shown in formula 1. The range of values for the DICE coefficient is 0 to 1, where 0 represents complete non overlap and 1 represents complete overlap, meaning that the segmentation result is completely consistent with the real segmentation. Therefore, the closer the DICE coefficient is to 1, the better the performance of the algorithm. The advantages of the DICE coefficient include its good robustness to imbalanced data, as it considers the size of intersections and unions.

$$\text{IoU} = \frac{\text{TP}}{\text{FP} + \text{TP} + \text{FN}} \quad (2)$$

As shown in formula 2. The range of values for IoU is 0 to 1, where 0 represents complete non overlap and 1 represents complete overlap. Therefore, the closer IoU is to 1, the more similar the segmentation results of the algorithm are to real segmentation, and the better the performance. IoU is one of the most commonly used evaluation indicators in image segmentation tasks, as it intuitively reflects the degree of overlap between the region generated by the algorithm and the real region. However, like other evaluation indicators, IoU also has its limitations, especially when dealing with imbalanced datasets, which may have some drawbacks. Therefore, other indicators such as DICE coefficient are usually combined to comprehensively evaluate the performance of the algorithm.

C. Results

Table 1. Unet and Swin Unet experimental results

Model	IoU	dice
Unet	0.81	0.889
Swin-Unet	0.83	0.910
Swin-Unet-EPA	0.84	0.915

From Table 1, it can be seen that Swin-unet performs better in segmentation than unet. The segmentation effect of Swin-Net-EPA is better than that of Swin-UNet. After adding EPA, the channel attention mechanism and spatial attention mechanism of EPA have added weights to some key input characteristics, resulting in better extraction of image features and improved segmentation accuracy.

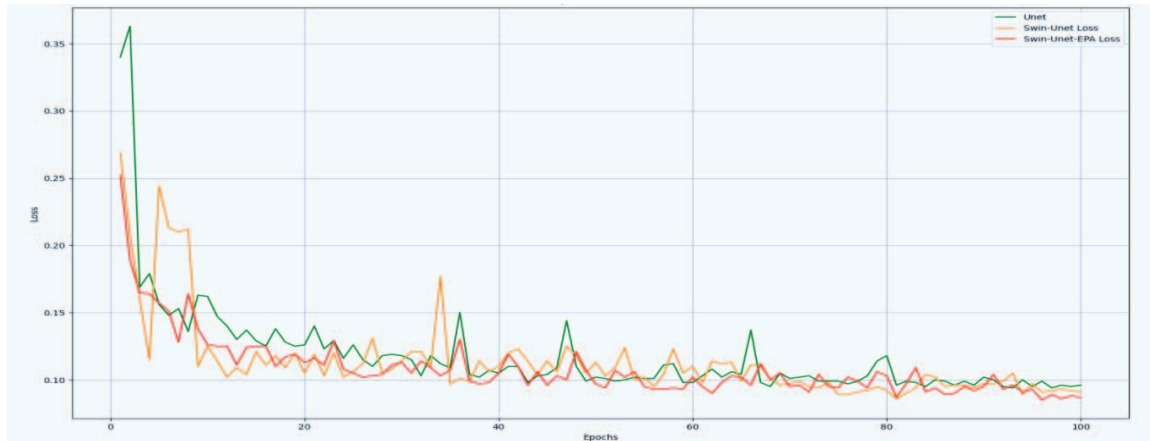


Figure 4. Loss

In Figure 4, we can clearly observe the training loss curve, which shows that the improved Swin-UNet-EPA method proposed in this paper quickly reached a lower loss value within the first 100 iterations and showed small fluctuations, gradually stabilizing. Compared with traditional Unet and Swin-UNet, the improved Swin-UNet-EPA network exhibits significantly lower loss values throughout the entire training process. This not only demonstrates the effectiveness of the MRI brain tumor automatic segmentation task, but also shows significant improvement compared to the original Swin-UNet network. Compared with other segmentation networks, our method also exhibits significant advantages. These results strongly support the excellent performance of the Swin-UNet-EPA network proposed in this paper in brain tumor segmentation tasks.



Figure 5. The segmentation results of different methods can be seen from the BraTs2021 test set. The first column is the ground truth. The second column is the Swin-UNet-EPA segmentation record, and the third column is the unet segmentation result.

From figure 5, Compared to Unet, the segmentation results of the Swin-UNet-EPA network are better in tumor region segmentation, and the edge contour segmentation results are also more accurate

V. Summarize

Swin-UNet combines U-Net and Swin Transformer. The segmentation effect of Swin Net is better than Unet, while the segmentation effect of Swin-UNet-EPA with EPA is better than that of Swin-UNet. U-Net typically has relatively few parameters and may be more suitable for use in situations with limited computing resources. The Swin Transformer may have more parameters, so in some cases it may require larger

computational resources. It is worth noting that this study still has some limitations and needs to be further explored in future work. In terms of experimental design, we noticed that the EPA in the model has been optimized to reduce the number of parameters in the model, but the Transformer still has huge parameters, which may have an impact on some aspects of the research. Future research will consider further exploring the segmentation of 3D brain tumor images to compensate for the shortcomings of this study in multimodality.

Acknowledgments

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